Energy Extension Questions

Extension: Energy of work done

1. When fully loaded, a fishing net has a mass of 500kg. Calculate the work done raising the fishing net 100m.

   Ew = 490000J (Ew = 490 KJ)

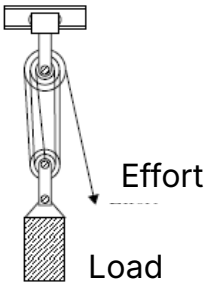
2. A weightlifter lifts a piece of gym equipment from the floor to arm's length, a vertical height of 1.2m. Calculate the mass of the equipment (in kg) if the minimum work done in lifting it is 600J.

   m = Ew/gd m =51kg

3. In lifting an engine out of a car, a mechanic uses a block and tackle. How much work is done if a force of 500N is used in pulling the rope a distance of 4m?

   Ew = 2KJ

4. A pulley lifting system designed to make it easier for an operator to lift a load onto a lorry is shown below. The operator applies an effort of 225N and pulls on 4.8m of rope to raise the load. Calculate the work done by the operator raising the load.



   Work = force x distance
   = 225 × 4.8
   = 1080 J

5. A mass of 50kg is raised to a height of 5m by a rope, which is wound around a pulley on a motor shaft of diameter 150mm. Determine the amount of work done by the motor.

Extension: electrical energy

1. A hot-air hand drier is activated for 30 seconds when the switch is pressed. The drier operates from a 230V supply and draws a current of 12A. Calculate the amount of electrical energy used when the drier is operating.

   E_e = Itv
   E_e = 12 × 30 × 230
   E_e = 82800 J
   E_e = 82.8 kJ

2. A 12 volt car battery has a capacity rating of 35 amp hours (that is, it can supply a current of 35 amps for 1 hour, or 5 amps for 7 hours, or any other combination giving an equivalence of 35 amp hours). Determine the amount of electrical energy the battery can supply.

   Ee = 1512000J

3. A portable electrical generator can deliver energy at a rate of 6kJ per second. Calculate the current that can be drawn from the generator if the electricity is supplied at a voltage of 110 volts.

   Ee = Ivt I = Ee/vt I = 54.55A

4. If the amount of electrical energy used by a 110V, 30A d.c. motor is 1.98MJ, for how long has the motor been in operation?

   Ee = Ivt t = Ee/Iv 600s

5. A turbine is used to drive a 110V electrical generator. If the output current is 3 amps, how much electrical energy does the generator produce in one hour?

   Ec = Ivt Ee = 1188KJ

6. An electric kettle takes 3.2 minutes to boil water. The original temperature of the water is 17°C and the kettle contains 1.8 litres of water. Calculate how much electrical current is used by the kettle from a 230 volt main supply.

   $E_H = cm\Delta T$
   $E_H = 4180 \times 1.8 \times (100-17)$
   $E_H = 1480 \times 1.8 \times 83$
   $E_H = 624492$ J
   $E_H = \underline{624.5\ kJ}$

   $E_e = Ivt$
   $624492 = I \times 230 \times (3.2mins \times 60)$
   $624492 = I \times 230 \times 192$
   $624492 = I \times 69{,}000$
   $I = 624492 / 44160$
   $I = \underline{14.14\Omega}$

7. The batteries on a prototype electric sports car is charged for 30 minutes from a main supply rated 230V 10A. Calculate the electrical energy used.

   $E_e = Ivt$
   $E_e = 10 \times 230 \times (30mins \times 60)$
   $E_e = 4140000$ J
   $E_e = 4140$ kJ
   $E_e = \underline{4.14\ MJ}$

Extension: kinetic energy

1. During a sheet-making process, 50kg ingots of metal are passed along rollers at a speed of 0.5 m/s. Calculate the kinetic energy of each ingot.

Ek = 6.25J

2. A racing car drives around a circuit at a speed of 50 m/s. If the car has a kinetic energy of 500kJ, what is the mass of the car?

   $E_K = ½mv^2$ m = E_k /1/2v^2 m = 400kg

3. A girl of mass 50 kg is riding on her bicycle and has a kinetic energy of 2.5kJ. What speed is the girl moving at, and what is the kinetic energy of the bicycle if it has a mass of 30kg?

4. A crane raises a load of 200kg a height of 30 m in one minute at uniform speed. Determine the kinetic energy of the load when it is moving.

   Ek = 58800J

5. An escalator has six people on it with a total mass of 900kg. If the escalator moves at a uniform speed of 0.5 m/s, what is the average amount of kinetic energy that each person contains?

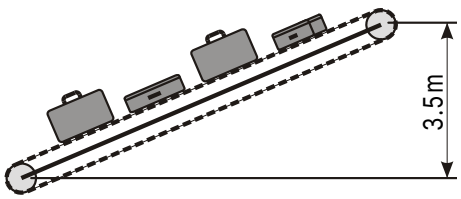
   Ek = 18.75J

6. Since 1975 the US government has required that a container carrying radioactive material must be able to withstand a high speed crash with an impact velocity of 250 m/s. By comparison, the UK safety crash standards (in 1994) are carried out at only 50 m/s. If the container has a mass of 10kg then calculate the percentage increase in kinetic energy of the American test over the British test.

   America: Ek = 312500J Britain: Ek = 12500J
   312500/12500 × 100% = 2500% increase

Extension: potential energy

1. Baggage handlers at an airport, place suitcases on to a conveyor belt, which lifts them up to the hold of the aeroplane as shown. Calculate the potential energy stored when a 20kg case is moved up the belt.



   Ep = mgh Ep = 20 × 9.8 × 3.5 Ep = 686J

2. At what height must a drum of mass 100kg be suspended above the ground if it possesses 4 kJ of potential energy?

   Ep = mgh h = Ep/mg h = 4000/ (100 × 9.8) h = 4.08m

3. Calculate the potential energy available from a reservoir holding 1800 litres of water at a height of 260m. (1 litre of water has a mass of 1kg.)

   Ep = mgh Ep = 1800 × 9.8 × 260 Ep = 4586400J

4. Metal piles are driven into the ground using a pile driver. The driver is raised to a height of 5 m above the ground and then released. Calculate the weight of the driver if the potential energy stored when it has been lifted is 9810 joules.

   Ep = mgh m = Ep/gh m = 9810/(9.8 × 5) m = 200.2kg

5. If a steeplejack has a potential energy of 2000J when he scales a ladder to a height of 10m, what amount of potential energy will he possess at a height of 15m?

   Ep = 3000J

Extension: heat energy

C_{water} = 4180J/kg/ºC C_{alum} = 899J/kg/ºC C_{copper} = 387J/kg/ºC
$C_{mercury}$ = 138J/kg/ºC

1. Calculate the heat energy required to heat 1.8kg of water from a temperature of 20ºC until it begins to boil.

   Eh = cmΔT Eh = 4180 × 1.8 × 80 Eh = 601920J

2. A hall has dimensions of 10 m × 328 m × 9 m. Assuming no heat losses, calculate the amount of heat energy required to raise the temperature of the air in the hall from 10ºC to 20ºC. The density of air is approximately 1.3 kg/m³ and the specific heat capacity is 850 J/kgK.

   m = 8 × 30 × 10 × 1.3 = 3120kg
   Eh = cmΔT Eh = 850 × 3120 × 10 Eh = 26520KJ

3. The heating element in an electric shower can produce heat energy at a rate of 7 kJ/s. Water enters the system at a temperature of 15ºC and is heated to 40ºC. Estimate the flow rate (in litres per second) and state any assumptions you have made.

   Eh = cmΔT m = Eh/cΔT m = 7000/(4180 × 25) m = 0.067 L/s

4. 54.3kJ of thermal energy is supplied to 1.3kg of oil having a specific heat capacity of 2.72kJ/kgK. If the initial temperature of the oil was 1.5ºC, what will be its final temperature?

5. During a test on an engine, the following data were recorded below. If it is estimated that the cooling jacket of water absorbs 25% of the energy supplied by the fuel, calculate the quantity of water required per second in the cooling jacket.
   - Oil consumption = 3 kg/hour
   - Specific heat capacity of oil = 1360 J/kgK
   - Temperature rise in cooling water = 20ºC

6. An electric heater supplies 13500J of heat energy to a metal block of mass 0.5kg. The temperature of the block rises from 20ºC to 80ºC during the heating process. Assuming that very little heat is lost from the block during the heating process, what is the specific heat capacity of the metal?

7. Calculate the energy required to raise the temperature of 750g of water from 34ºC to 91ºC.

8. When 6.3×10^5J of heat energy is added to a statue in a kiln oven, the statue's temperature rises from 16 ºC to 95 ºC. The specific heat capacity of the statue material is 3500J/Kg/C. Calculate the mass of the statue. Calculate the final temperature of 120g of water when 2.4×10^4J of heat is added to it. The initial temperature of the water is 19ºC.

9. When 3kg of water is cooled from 80ºC to 10ºC how much heat energy is lost?

10. How much heat is needed to raise a 0.30kg piece of aluminium from 30ºC to 150ºC?

11. Calculate the temperature change when 10.0kg of water loses 232kJ of heat.

12. Calculate the temperature change when 1.96kJ of heat is added to 500g of copper.

13. 2.52×10^4J of heat is added to 2kg of mercury to reach a final temperature of 130ºC. What was the initial temperature of the mercury?

Extension: power

1. A winch lifts a 500kg pallet of bricks to a height of 10m in a time of 15 seconds. Calculate the minimum output power of the lift.

   $E_P = mgh$
   $E_P = 500 \times 9.8 \times 10$
   $E_P = 49{,}000$ J
   $E_P = 49$ kJ

   $P = E/t$
   $= 49{,}000/ 15$
   $= 3266.7$ watts

2. An electric heater is rated at 3kW. Calculate how much heat energy the heater will deliver in one hour.

   $P = E_H/t$
   $3{,}000 = E_H$ / (60 mins x 60 seconds)
   $E_H = P \times t = 3000 \times 3600$
   $E_H = 10.8$ MJ

3. A car of mass 800kg is moving at a steady rate of 70km/hour. Calculate the kinetic energy of the car and the power output from the car in one minute.

   m = 800kg v = 70km/hr = 19.4ms^{-1} $E_k = 1/2mv^2$ t = 60s
   $E_K = ½mv^2$ Ek = ½ x 800 × 19.4^2 = 150544J

P = E/t
P = 150544 / 60
P = 2509.067W

4. An electric motor is rated at 10kW and runs from a main supply of 220V. If it consumes 10kJ of energy over a period of 30 seconds, what is the current used by the motor?

V = 220v Ee = 10 000J t = 30s
Ee = Ivt I = Ee / vt I = 10 000 / (220 × 30) = 1.52A

Extension: calculating energy transfers

1. A body of mass 30kg is projected vertically upwards with an initial velocity of 20 m/s. What is the initial kinetic energy of the body and to what height will it rise?

F = mg $E_K = ½mv^2$ Ek = ½ x 30 × 20^2= 6000J
Ep = mgh 6000 = 30 × 9.8 x h h = Ep / mg h = 6000 / (30 × 9.8)
h = 20.41m

2. In a stamping machine, the die has a mass of 35kg and falls through a height of 2m on to a metal block. If the depth of indentation is 10 mm, find the average stamping force. Assume no rebound.

3. A boy stands on a skateboard at the top of a steep hill that has a height of 500m and allows gravity to increase his speed all the way to the bottom of the hill. If the boy has a mass of 60kg, calculate his potential and kinetic energy at the top, middle and bottom of the hill. (Ignore friction or any other resistance.)

Extension: efficiency

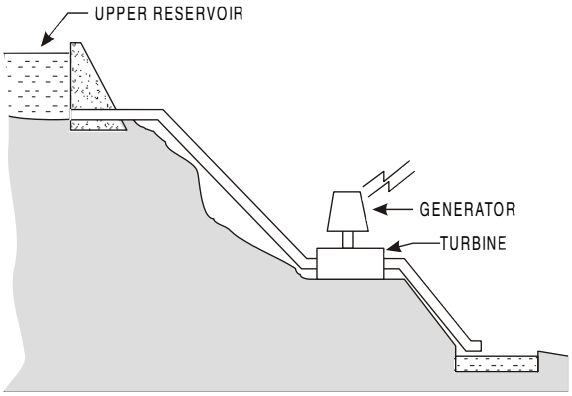
1. An electric kettle is rated at 240V, 10A. When switched on it takes three minutes to raise the temperature of 0.5kg of water from 20ºC to 100ºC. Determine:

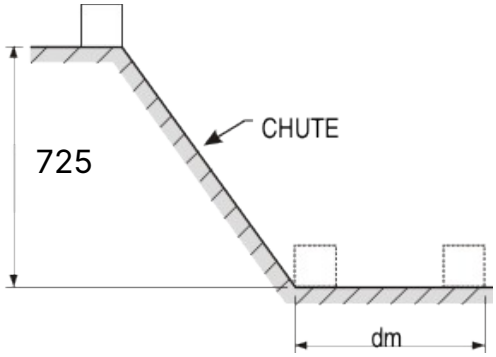
(a) the electrical energy supplied in the three minutes

(b) the heat energy required to raise the temperature of the water

(c) the efficiency of the kettle.

2. In a hydroelectric electricity generating station, water is allowed to flow downhill through a turbine, which is connected to a generator. The water falls through a vertical height of 725m at a rate of 5,125kg/s. If the energy transfer is 72% efficient, determine the amount of electrical energy produced per second.





E_P = mgh
= 5125 × 9.81 × 725
= 36450281.25
= 36.5 MJ

72% = 26.2 MJ

3. Boxes in a factory are transferred from one floor to another using a chute system as shown above. The boxes start from rest at the top of the chute and during the decent there is a 40% loss of energy. The boxes weigh 10kg each.

a) Calculate the velocity of the boxes at the bottom of the chute.

b) What is the distance *d* that each box will travel along the bottom floor before coming to rest if the frictional force opposing the motion of each package is 25% of its weight?

Extension: energy audits

1. A deep fat fryer uses 2kJ of electrical energy per second. It can raise the temperature of 1.5kg of cooking oil from 2°C to 80°C in 6 minutes. If the specific heat capacity of the oil is 2400 J/kgK, calculate the heat energy taken in by the oil every second and hence the efficiency of the fryer.

2. When full, a container of water holds 1.5 litres of water. The initial temperature of the cold water is 20°C. A heater that operates from a 240V supply draws a current of 8A and has an efficiency of 75%. Determine how long it will take to heat the container of water to 90°C.

3. A lift is used to raise a mass of 200kg to a height of 10m in two minutes. If the lift motor produces 12 MJ of energy in one hour, how efficient is the lift?

4. A vending machine, which dispenses hot drinks, heats up one cupful of water (0.15kg)

from 30ºC to 90ºC. The heating element operates from a 240 volt supply and has a resistance of 12 ohms.

(a) Calculate the current drawn by the heating element.

(b) Calculate the heat energy transferred to the water.

(c) Calculate the time taken to heat the water if the system is 100% efficient.

(d) Calculate the percentage of energy lost if it actually takes 10 seconds to heat the water.

(e) Draw a system diagram showing the input to and outputs from the system. Include the percentages.